CLASSIFICATION

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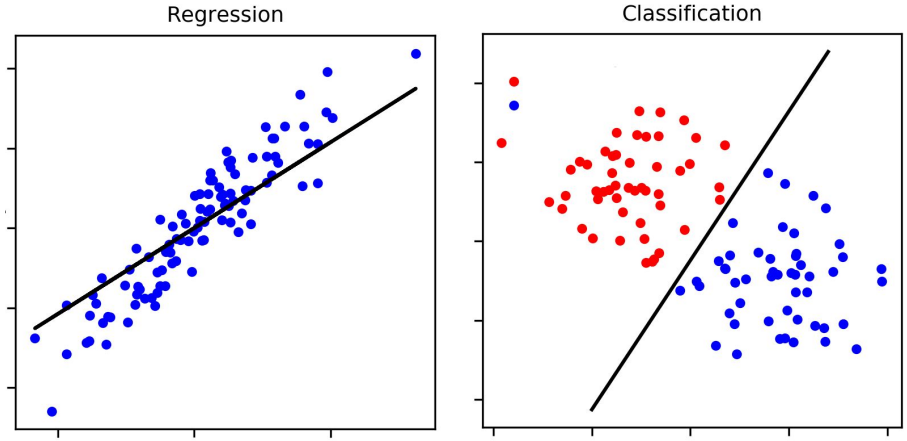
Classification

- A supervised ML task in which the model predict the categorical outcome
- Classification task can be further divided into:
  - Binary or two class classification
  - Multiclass classification

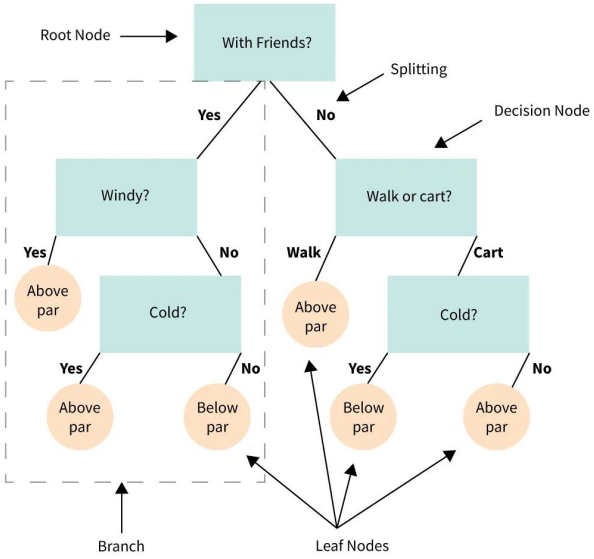


Classification algorithms

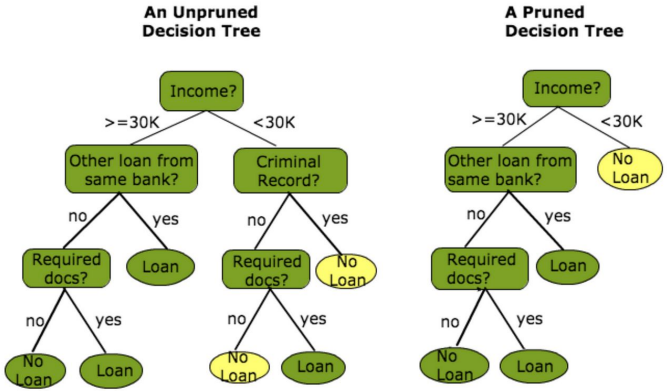
- Just to list a few:
  - Logistic Regression
  - Decision Trees
  - Random Forest
  - Support Vector Machines (SVM)
  - k-Nearest Neighbors (kNN)
  - Naive Bayes
  - Artificial Neural Networks (ANN)
- Full list of algorithms in parsnip package

Decision tree

- Can be used for regression and classification - classification and regression trees (CART) models
- The order of the variable to be splitted is determined by the purity:
  - Gini impurity
  - Entropy and information gain
- Purity - how well separated the points are at the nodes
- The split in decision tree for classification aims to maximise the separability of the class (i.e., impurity)

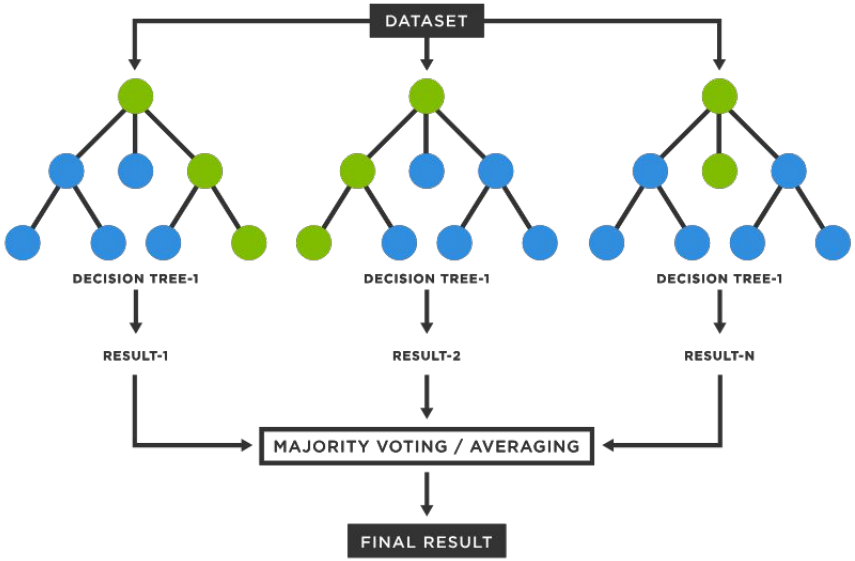


- Pros:
  - Easy to understand
  - Fast computation
  - Able to handle missing data and outliers
- Cons:
  - Tree overfitting - can be overcome with a pruning or CV methods



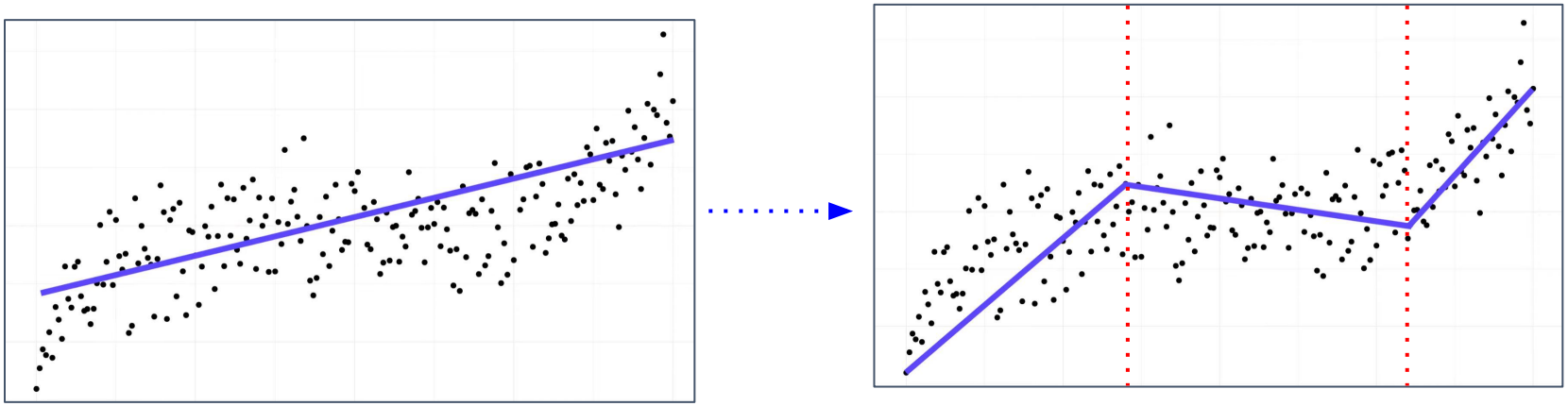
Random forest

- Basically, a collection of decision tree
- Pros:
  - Low risk of overfitting
  - Usually more accurate
  - Able to handle missing data and outliers
- Cons:
  - Relatively slow computation
  - Low interpretability



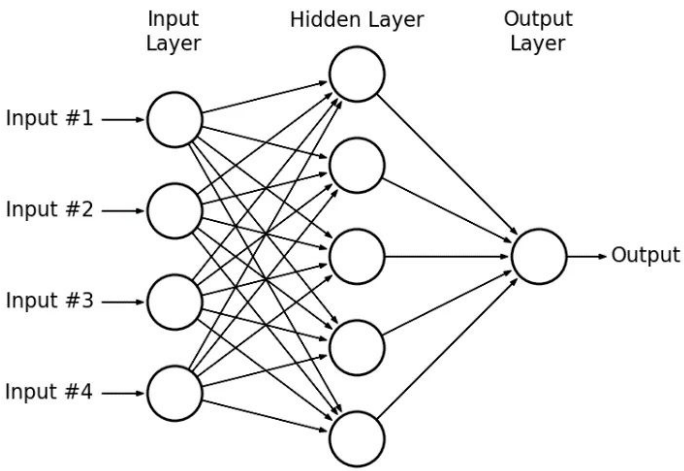
MARS

- Multivariate adaptive regression splines - MARS
- Non-parametric regression technique
- Introduced in 1991 by Friedman
- Main idea - cut the regression line into several cut-off points

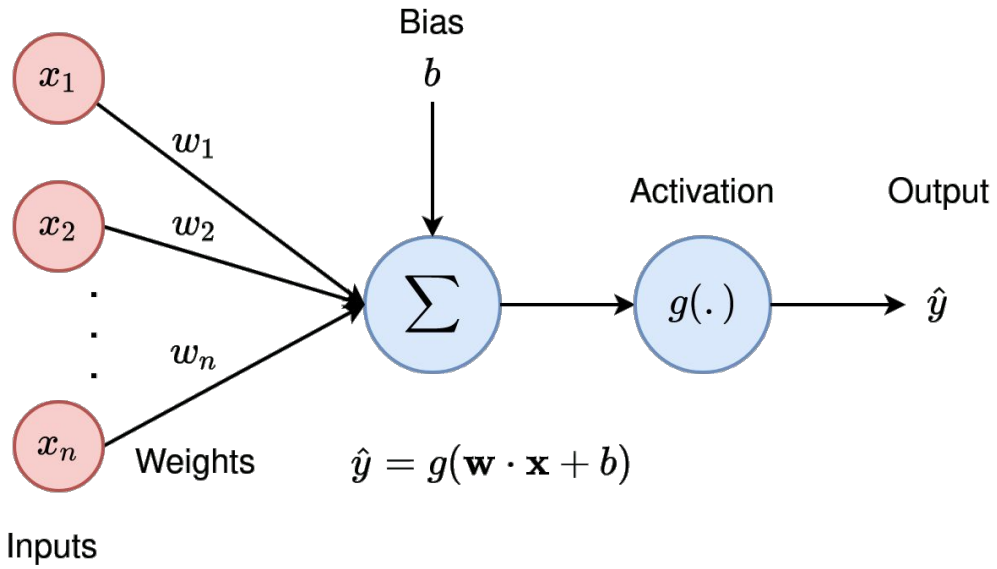


Artificial Neural Network (ANN)

- Neural network is the basis of deep learning
- ANN can be used for regression and classification
- Simple ANN is a single layer, feed-forward neural network, which also known as multilayer perceptron (MLP)
- ANN is formed of:
  - Input layer
  - Hidden layer
  - Output layer



- Breakdown of ANN:



Performance metrics

Confusion matrix

- It is a table comparing the predicted and the actual classes
- Best confusion matrix is at a [wiki page]

		Predicted Class: Positive	Predicted Class: Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	**Sensitivity** $\frac{TP}{(TP+FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	**Specificity** $\frac{TN}{(TN+FP)}$
		Precision $\frac{TP}{(TP+FP)}$	**Negative Predictive Value** $\frac{TN}{(TN+FN)}$	**Accuracy** $\frac{TP+TN}{(TP+TN+FP+FN)}$

Accuracy

- Proportion of correctly classified cases out of the total cases

Sensitivity/recall

- High sensitivity means effective at detecting the true positive cases
- High sensitivity means low false negative

Specificity

- High specificity means effective at detecting the true negative cases
- High specificity means low false positive

Precision/positive predictive value (PPV)

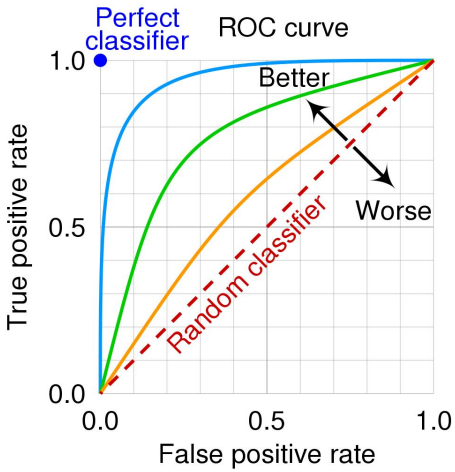
- Indicates proportion of subjects with a predicted positive who truly positive
- High precision - low false positive

Negative predictive value (NPV)

- Indicates proportion of subjects with a predicted negative who truly negative
- High NPV - low false negative

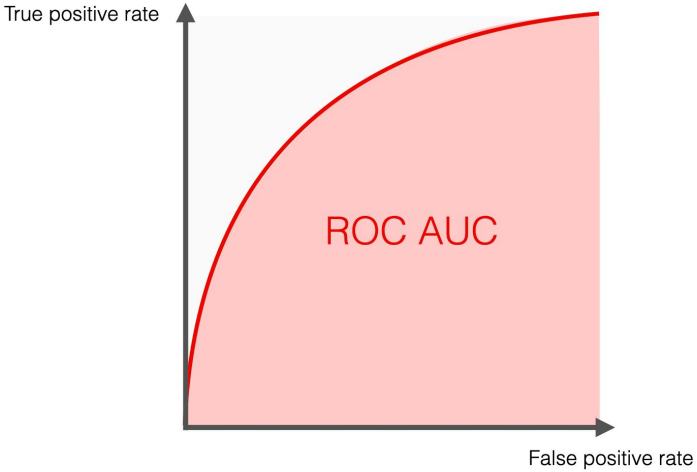
Receiver operating characteristic (ROC) curve

- Reflects a performance of classification models at certain threshold (usually 0.5)
- Can be used to compare different classification ML models



ROC-Area under the curve (ROC-AUC)

- It provides an aggregate measure of the model's performance
- AUC of 0.5 = no discriminant ability, while AUC of 1 = perfect classification
- Can be used to compare different classification ML models



Suggested readings/references

- Kuhn, M., & Silge, J. (2022). Tidy Modeling with R: A Framework for Modeling in the Tidyverse. O'Reilly Media.
- Burger, S. V. (2018). Introduction to machine learning with R: Rigorous mathematical analysis (First edition). O'Reilly Media.

Any question?

Hands-on in R

Data - Pima Indians Diabetes

- Aims - to predict whether or not a patient has diabetes or not
- Data source is here
- All patients are females at least 21 years old of Pima Indian heritage
- Variables:
  - Pregnant - number of times pregnant
  - Glucose - plasma glucose concentration (glucose tolerance test)
  - Pressure - diastolic blood pressure (mm Hg)
  - Triceps - triceps skin fold thickness (mm)
  - Insulin - 2-Hour serum insulin (mu U/ml)
  - Mass - Body mass index (weight in kg/(height in m)
  - Pedigree - Diabetes pedigree function
  - Age - Age (years)
  - Diabetes - Class variable (test for diabetes)

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